Survival on a Tropical Island: student guide

The project page is at whatthecell.org/csp-survival-island.html

IB Diploma Programme | Collaborative Sciences Project

Survival on a Tropical Island

Student Guide: Your Team's Roadmap

THIS GUIDE WALKS YOU AND YOUR TEAM THROUGH EVERYTHING YOU NEED TO DO OVER THE NEXT TWO AND A HALF DAYS. KEEP IT WITH YOU (OR KEEP IT OPEN) AT ALL TIMES — CHECK OFF EACH TASK AS YOU FINISH IT, AND USE IT TO MAKE SURE YOUR TEAM IS ON TRACK FOR EVERY DEADLINE.

REMEMBER: THIS PROJECT IS NOT GRADED ON HOW POLISHED YOUR PRESENTATION LOOKS. IT'S GRADED ON YOUR PROCESS — HOW WELL YOU COLLABORATED, COMMUNICATED, AND THOUGHT SCIENTIFICALLY AS A TEAM. YOUR ONLY INDIVIDUALLY ASSESSED PIECE OF WORK IS YOUR 100-WORD REFLECTION ON THE LAST AFTERNOON.

Your Mission

A fast-moving virus has collapsed global civilization. You and your team are among the survivors. You do not carry the disease. You have a small boat and one chance to gather supplies before leaving the mainland forever.

Choose a real island in Southeast Asia or the Pacific. Research it. Then, as a team, design a complete colony that can sustain your community — and future generations born on the island — indefinitely.

You must address, as a team:

- Food, water, shelter, energy, health, waste, governance, and population

Your constraints:

- Everything you bring must fit in a small boat (approx. 6 metres / 20 feet, roughly 1,500–2,000 kg)
- Your community must be large enough to be genetically and socially self-sustaining
- Your colony must not destroy the island's ecology to survive

Your Team

Fill this in at the start of Day 1. Every team needs at least one student thinking about each of these four areas — write in who is taking the lead on each, even if everyone contributes to everything.

NAME	GRADE	SUBJECT	ROLE IN THE COLONY

Our chosen island: _____

Our starting population size: _____

Key Deadlines at a Glance

- ☐ ISLAND RESEARCH DOSSIER — DUE END OF DAY 1
- ☐ ANNOTATED SUPPLIES MANIFEST — DUE HALFWAY THROUGH DAY 2 (MIDDAY)
- ☐ SYSTEMS DIAGRAM & INTERDISCIPLINARY CONNECTIONS — DUE END OF DAY 2
- ☐ COLONY DESIGN JOURNAL — KEEP UPDATING IT EVERY DAY, ALL THREE DAYS
- ☐ COLONY PITCH PRESENTATION — DELIVERED DAY 3 MORNING (OCTOBER 7TH)
- ☐ INDIVIDUAL 100-WORD REFLECTION — WRITTEN AND SUBMITTED DAY 3 AFTERNOON

Day 1 — Scenario, Research & Supplies Debate

By the end of today: your island is chosen and researched, and you've started debating what to bring.

Session 1 — Launching the Scenario

- ☐ Listen to the scenario briefing and ask questions if anything is unclear
- ☐ Take part in the map activity — mark where you'd go and why
- ☐ Meet your team and find out who's researching what

Session 2 — Choosing Your Island & Starting Research

- ☐ As a team, agree on your island (use Google Earth, GBIF, and the IUCN database to check it has enough data)
- ☐ Open your shared Colony Design Journal and record your island choice and why you chose it
- ☐ Split up research by subject and start investigating your area:
- ☐ ESS — ecosystem, land area, climate, rainfall, carrying capacity
- ☐ Biology — edible species, disease risks, freshwater life, soil ecology
- ☐ Chemistry — water sources and quality, soil chemistry, minerals
- ☐ Physics — topography, wind, sun exposure, rivers, building materials

Session 3 — Finishing Your Research & the Supplies Debate

- ☐ Keep researching until your Island Research Dossier is complete in the journal
- ☐ Join the Supplies Debate — each person argues for the items their subject says are most important
- ☐ Record every item discussed in the journal, including ones you rejected and why

DEADLINE — END OF DAY 1: ISLAND RESEARCH DOSSIER

MAKE SURE YOUR JOURNAL INCLUDES: GEOGRAPHY, CLIMATE, RESOURCES, AND HAZARDS FOR YOUR ISLAND, PLUS INITIAL FINDINGS FROM ALL FOUR SUBJECTS. YOUR TEAM WILL SHARE YOUR ISLAND CHOICE AND TOP 5 SUPPLIES WITH THE GROUP BEFORE YOU FINISH TODAY.

Ask yourselves before you finish Day 1:

- Can we actually find good data on this island? If not, is it worth switching now?
- Have we backed up our top 5 supply choices with real evidence, not just 'common sense'?

Day 2 — Supplies, Colony Systems & Presentation Building

By the end of today: your supplies list is locked in, your colony systems are designed, and your presentation is drafted.

Session 4 — Finalising Your Supplies Manifest

☐ Finalise your full list of supplies — every single item needs a written reason

☐ Double-check your total weight stays within 1,500–2,000 kg

☐ Be ready to explain: what would you do without this item? Why is it worth the space?

DEADLINE — HALFWAY THROUGH DAY 2 (MIDDAY): ANNOTATED SUPPLIES MANIFEST

SUBMIT YOUR FINALISED, ANNOTATED SUPPLIES MANIFEST TO YOUR TEACHER AT THE MIDDAY CHECKPOINT. BE READY TO SHARE YOUR FINAL POPULATION SIZE AND ONE HARD TRADE-OFF YOUR TEAM HAD TO MAKE.

Session 5 — Designing Your Colony Systems

☐ Each person designs their subject's core system and records it in the journal

☐ Draw a systems diagram showing how your colony's systems connect and depend on each other

☐ Look for gaps — missing waste plan? No backup food source? Fix them now, not on Day 3

☐ Identify at least 3 clear places where two or more subjects had to work together

DEADLINE — END OF DAY 2: SYSTEMS DIAGRAM & INTERDISCIPLINARY CONNECTIONS

SUBMIT YOUR COMPLETED SYSTEMS DIAGRAM AND YOUR LIST OF AT LEAST 3 INTERDISCIPLINARY CONNECTIONS, ALONG WITH AN UPDATED JOURNAL NOTING ANY GAPS YOU FOUND AND HOW YOU FIXED THEM.

Session 6 — Building Your Presentation

☐ Choose your format: slides, poster, whiteboard, or a combination

☐ Each person builds the section for their own subject expertise

☐ Assign who speaks when — every team member should have a speaking role

☐ Show, don't tell: use a map, a diagram, or a chart wherever you can instead of bullet points

☐ Aim to leave today with a complete draft — finish any polishing before tomorrow morning

Ask yourselves before you finish Day 2:

- If someone challenged us on any single system right now, could we defend it?
- Does our presentation tell one connected story, or does it feel like four separate reports?

Day 3 (October 7th) — Presentations & Reflection

Half day. This morning you pitch your colony. This afternoon you reflect on the experience.

Session 7 — Final Preparation

- ☐ Use these 30 minutes for last adjustments, printing, or tech setup
- ☐ Do one quick run-through as a team if you have time

Session 8 — Colony Pitch Presentations

- ☐ Deliver your 15-minute Colony Pitch — remember, every team member speaks
- ☐ Handle your 5-minute challenge round — it's fine not to know an answer; say what you'd research next
- ☐ Listen to other teams present, and ask genuine questions when it's your turn to challenge

DEADLINE — DAY 3 MORNING: COLONY PITCH PRESENTATION

YOUR 15-MINUTE COLONY PITCH PLUS THE 5-MINUTE CHALLENGE ROUND HAPPENS THIS MORNING. YOU ARE CONVINCING YOUR AUDIENCE — PROSPECTIVE COLONISTS — THAT YOUR COLONY DESIGN WILL GENUINELY WORK.

Session 9 — Debrief & Your Individual Reflection

- ☐ Take part in the whole-group debrief discussion
- ☐ Write your own 100-word reflection, independently — this is not a team task
- ☐ Save and submit your reflection electronically before you leave

DEADLINE — END OF DAY 3, AFTERNOON: INDIVIDUAL REFLECTION

THIS IS YOUR ONLY INDIVIDUALLY ASSESSED PIECE OF CSP EVIDENCE. WRITE AND SUBMIT IT BY THE END OF THE AFTERNOON SESSION.

YOUR REFLECTION: WHAT TO WRITE ABOUT

IN ABOUT 100 WORDS, BE HONEST ABOUT YOUR EXPERIENCE. CONSIDER:

- WHAT WAS IT LIKE INTEGRATING YOUR SCIENTIFIC KNOWLEDGE WITH THREE OTHER SUBJECTS?
- WHERE DID DISAGREEMENTS IN YOUR TEAM COME FROM, AND HOW DID YOU RESOLVE THEM?
- WHAT DID YOU LEARN ABOUT SUSTAINABILITY THAT YOU COULDN'T HAVE LEARNED FROM YOUR OWN SUBJECT ALONE?
- WHAT WOULD YOU DO DIFFERENTLY — IN THE DESIGN, OR IN HOW YOUR TEAM WORKED TOGETHER?

BE SPECIFIC. DESCRIBE ONE REAL MOMENT — NOT A POLISHED SUMMARY OF WHAT YOUR TEAM PRODUCED.

Quick Reference

Your Role by Subject

SUBJECT	YOU ARE THE…	YOU'RE THE GO-TO PERSON FOR
ESS	SUSTAINABILITY & LAND-USE PLANNER	ECOSYSTEM RESEARCH, CARRYING CAPACITY, LAND USE, WASTE AND ENVIRONMENTAL RISK
BIOLOGY	FOOD, HEALTH & POPULATION OFFICER	EDIBLE SPECIES, FOOD PRODUCTION, DISEASE PREVENTION, SANITATION, POPULATION SIZE
CHEMISTRY	WATER, SOIL & MATERIALS SPECIALIST	FRESHWATER PURIFICATION, SOIL CHEMISTRY, MATERIALS, CHEMICAL SAFETY
PHYSICS	INFRASTRUCTURE & ENERGY ENGINEER	SHELTER DESIGN, RENEWABLE ENERGY, WATER ENGINEERING, CONSTRUCTION

Colony Pitch: What to Cover

- Hook & island introduction
- Island profile (ESS)
- Your community: size and governance (All)
- Food systems (Biology)
- Water & soil (Chemistry)
- Shelter & energy (Physics)
- Health & sanitation (Biology/Chemistry)
- Supplies manifest highlights (All)
- Interdisciplinary connections (All)
- Long-term sustainability (ESS)
- Your biggest risk and your plan (All)

Challenge Round: Questions You Might Get

- Food: What do you eat in Week 1, before any crops are ready?
- Water: How exactly does water get from the source to a safe drink?
- Health: Someone breaks a leg in Month 2 — what happens?
- Governance: Two groups disagree about food distribution — how is it resolved?
- Energy: Your solar panels degrade in 10–15 years — then what?
- Sustainability: Your population triples in 20 years — is the island still viable?
- Resilience: A cyclone destroys half your shelters and crops — does the colony survive?

A FEW REMINDERS

- YOU DO NOT NEED IMAGES OF THE REAL ISLAND — HAND-DRAWN MAPS AND DIAGRAMS WORK JUST AS WELL
- EVERY TEAM MEMBER SHOULD SPEAK DURING THE PRESENTATION
- IF YOU CAN'T ANSWER A CHALLENGE QUESTION, SAY WHAT YOU'D RESEARCH NEXT — THAT'S A PERFECTLY GOOD ANSWER
- KEEP YOUR COLONY DESIGN JOURNAL UPDATED EVERY DAY — IT'S YOUR EVIDENCE OF THE WHOLE PROCESS

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